

Unmasking the Board.

Behavioral anomaly detection in online chess, an unsupervised triage system for human moderators.

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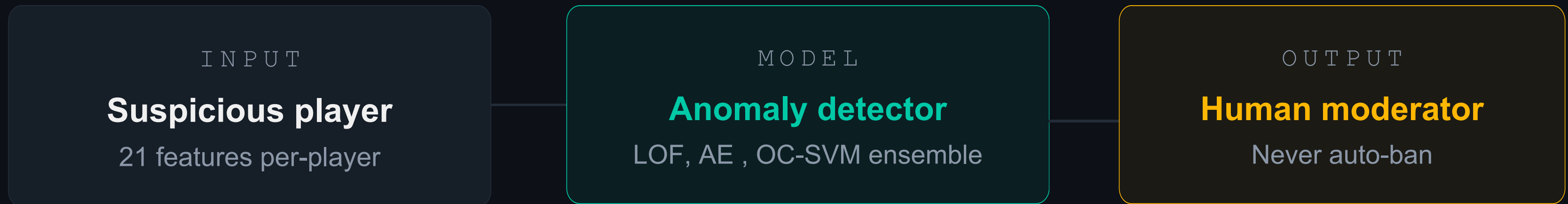
Chess cheating has no ground truth.

Engine assistance leaves **statistical fingerprints** such as win rates, game length, opening depth.

Platforms **never publish** confirmed cheater lists.

No labels → **supervised learning is impossible.**

Only valid path: **unsupervised anomaly detection.**



Data and Preprocessing

500k games, 17,909 players, one leakage fix.

6.25M

Lichess games available (July 2016 export)

500k

Sampled: rapid + classical only

17,909

Unique players ≥ 5 games

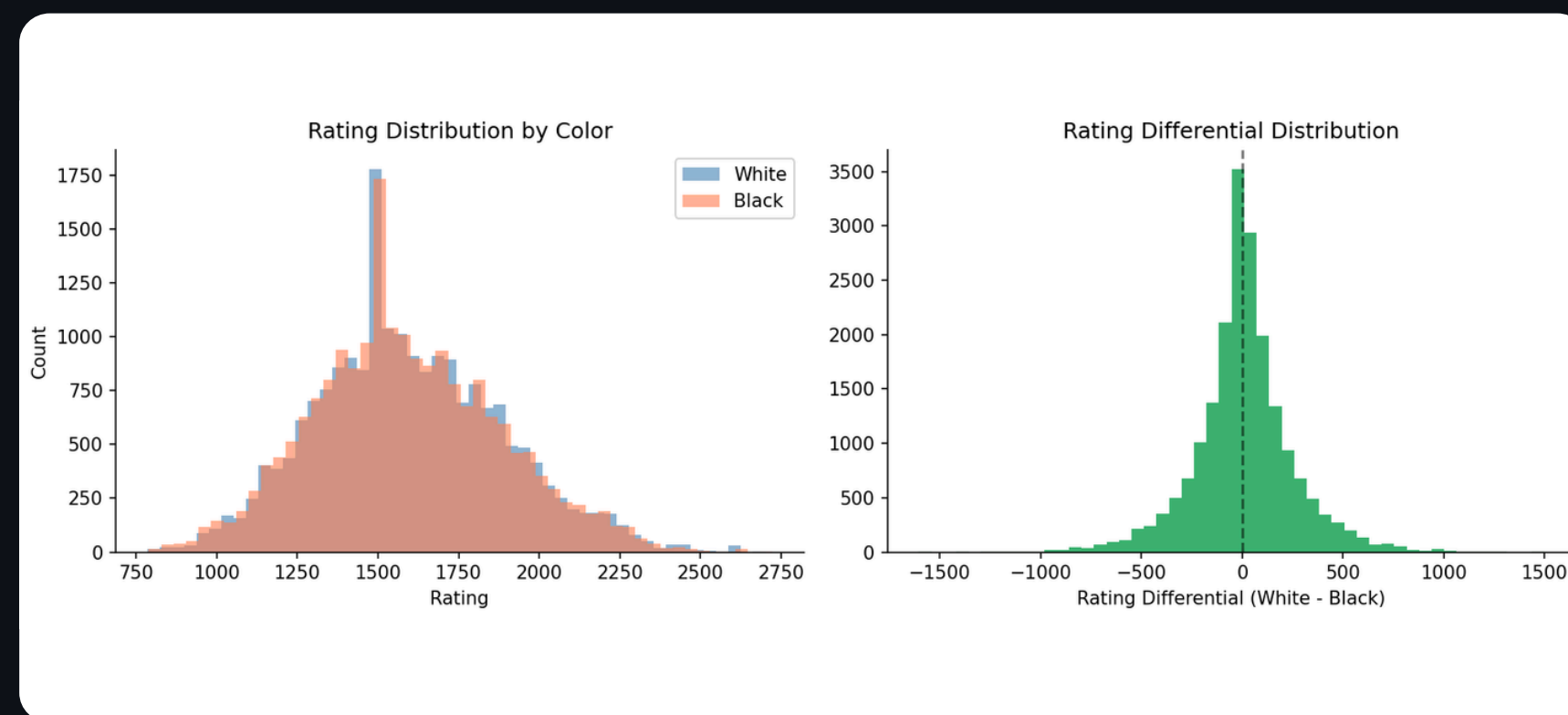
12,738

Flagging-eligible ≥ 15 games

Blitz excluded as engine use requires **thinking time** ; blitz signal is noise.

70 / 15 / 15 player-level split *before* any normalization.

StandardScaler fit on training players only; rating-band z-scores recomputed from training to avoid leakage.



Rating distribution centered around 1500.

Feature Engineering

21 features: behavior + chess accuracy.

Behavioral signals

8 features

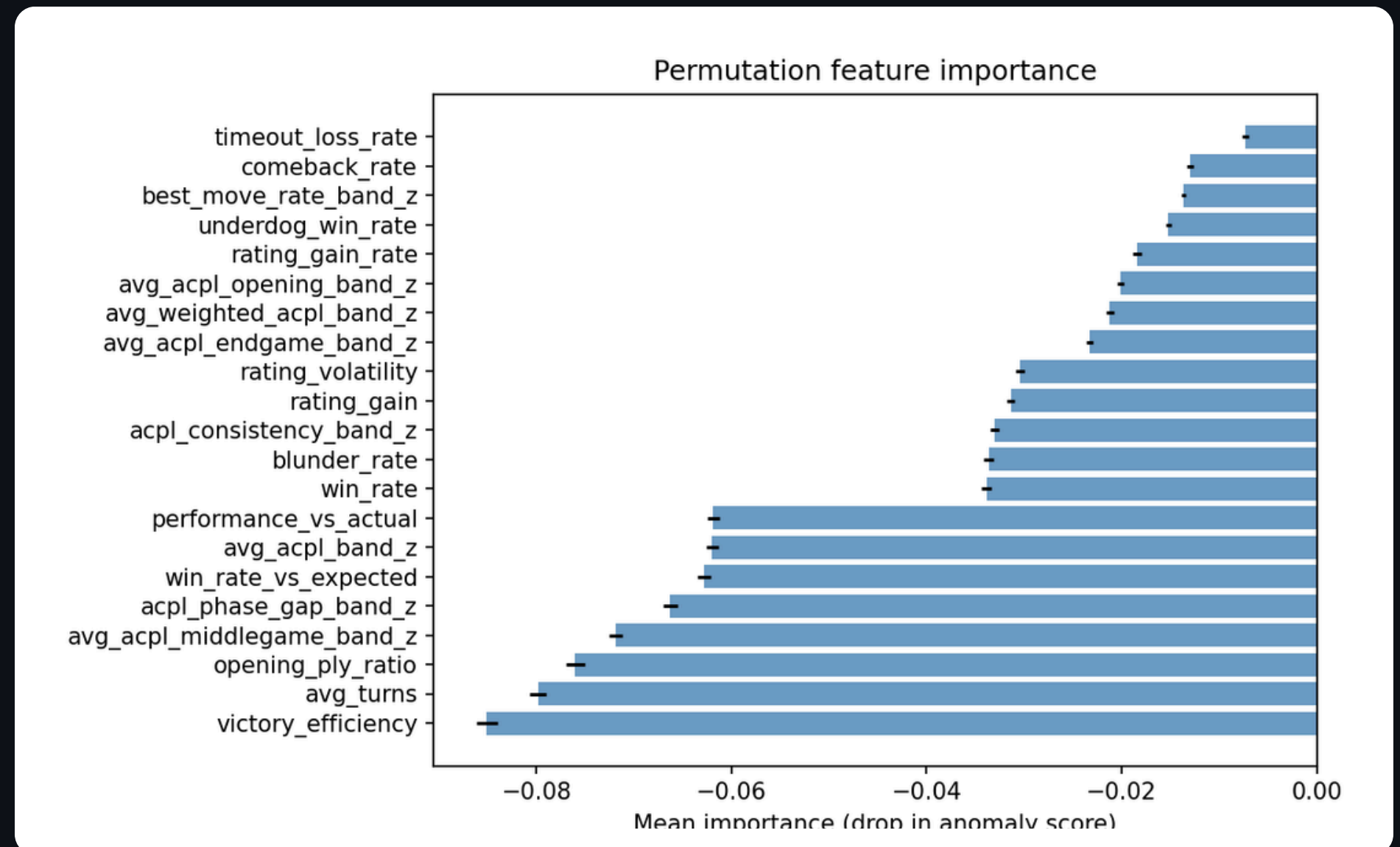
win_rate_vs_expected
victory_efficiency
opening_ply_ratio
rating_volatility

Stockfish ACPL

13 features

avg_acpl_{opening,mid,end}_band_z
acpl_phase_gap_band_z

All accuracy features band-normalized within **9 Elo tiers**.



Top 3 by permutation importance: **victory_efficiency**, **avg_turns**, **opening_ply_ratio**.

Models

Seven detectors, three algorithmic families.

MODEL	TYPE	
Z-Score Baseline	Statistical threshold	baseline
Isolation Forest	Tree ensemble	excluded
LOF ★	Density-based	BEST
One-Class SVM	Kernel boundary	in ensemble
Autoencoder	Neural reconstruction	in ensemble
ACPLSubAutoencoder	Diagnostic variant	accuracy-only
HDBSCAN	Clustering	excluded

PRODUCTION ENSEMBLE

Top 3 by CV AUC LOF +
Autoencoder + OC-SVM

Majority vote, ≥ 2 of 3 → **flagged** All 3
agree → **high-confidence shortlist**

Z-Score retained as transparent baseline; Isolation
Forest & HDBSCAN excluded for low AUC.

Evaluation Framework

No labels → synthetic anomaly injection.

engine_perfect

All features pushed to the 99th percentile .

PRIMARY

subtle

One third of features shifted $\pm 1.5\sigma$. Stays on the data manifold as the real benchmark.

realistic_cheater

Low ACPL only, normal behavioral features.
Mimics evasion and tests the hard ceiling.

5-fold CV on training split.

METRICS

ROC-AUC

Avg.
Precision

Recall@k

Results

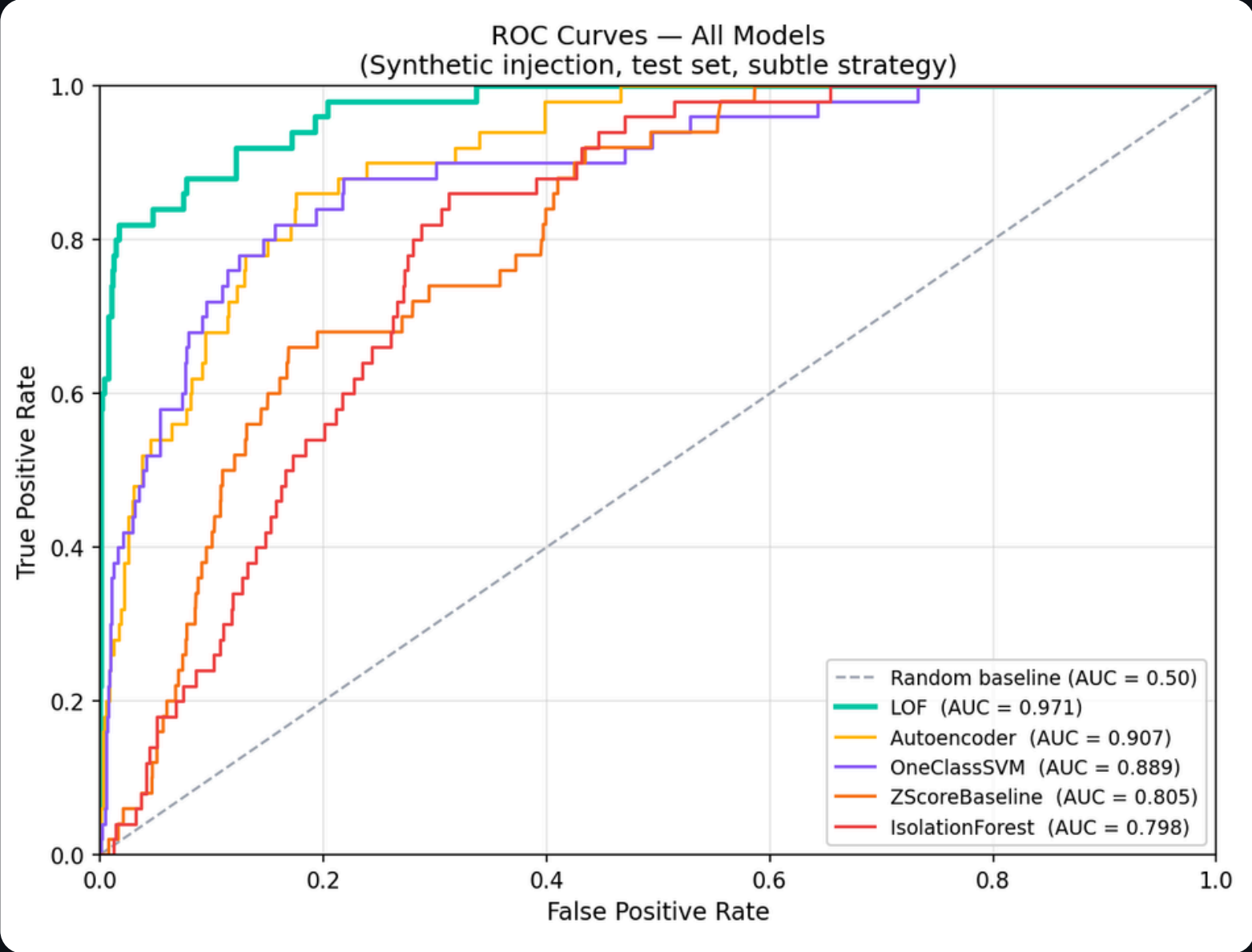
LOF wins by a large margin.

312/17,909

MODEL	CV ROC-AUC	TEST AUC
LOF ★	0.959 ± 0.031	0.971
Autoencoder	0.893 ± 0.023	0.907
One-Class SVM	0.877 ± 0.021	0.890
Z-Score Baseline	0.780 ± 0.031	0.805
Isolation Forest	0.760 ± 0.016	0.798
HDBSCAN	0.524 ± 0.054	0.625

LOF, test ROC-AUC

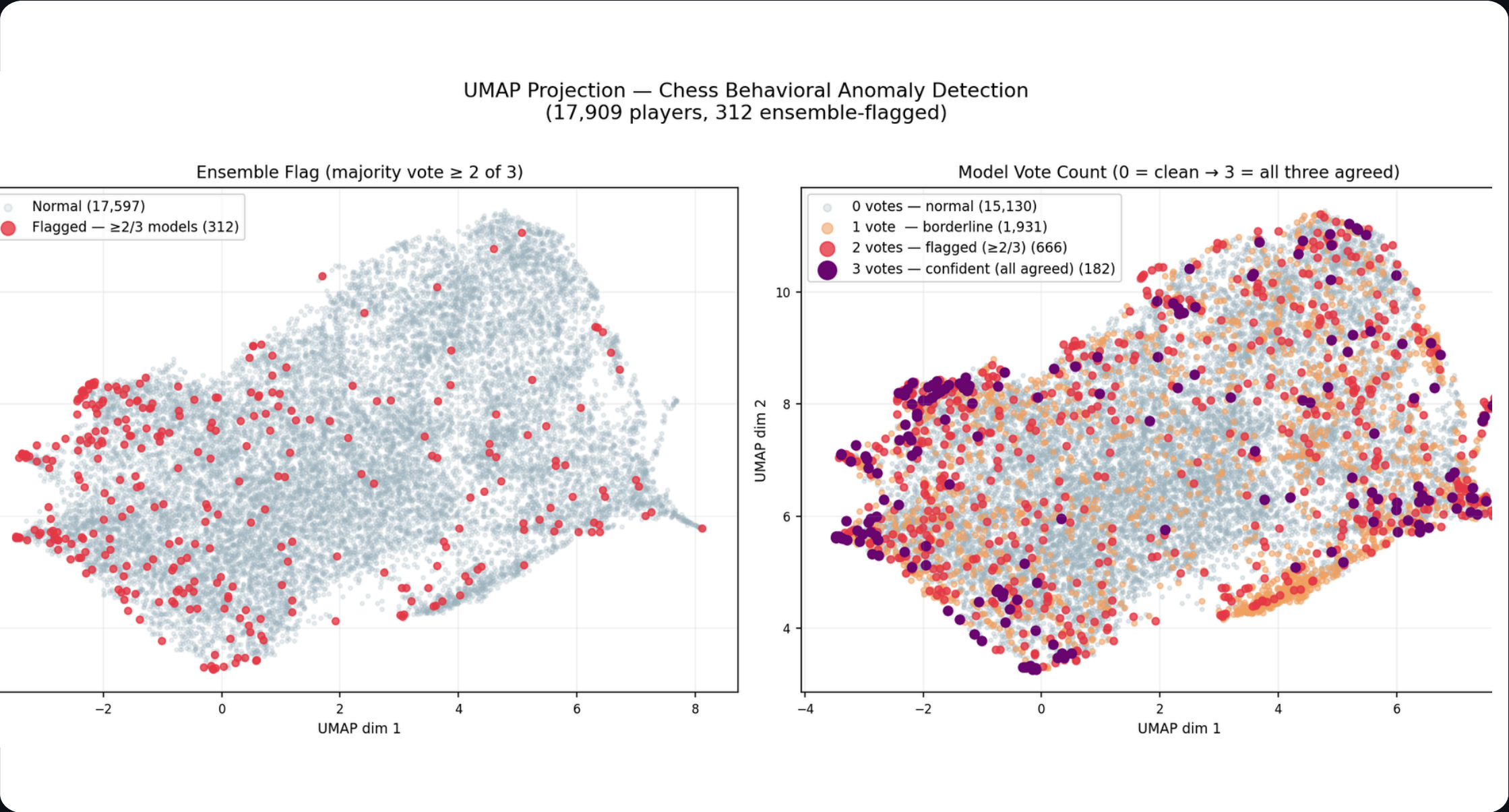
0.971



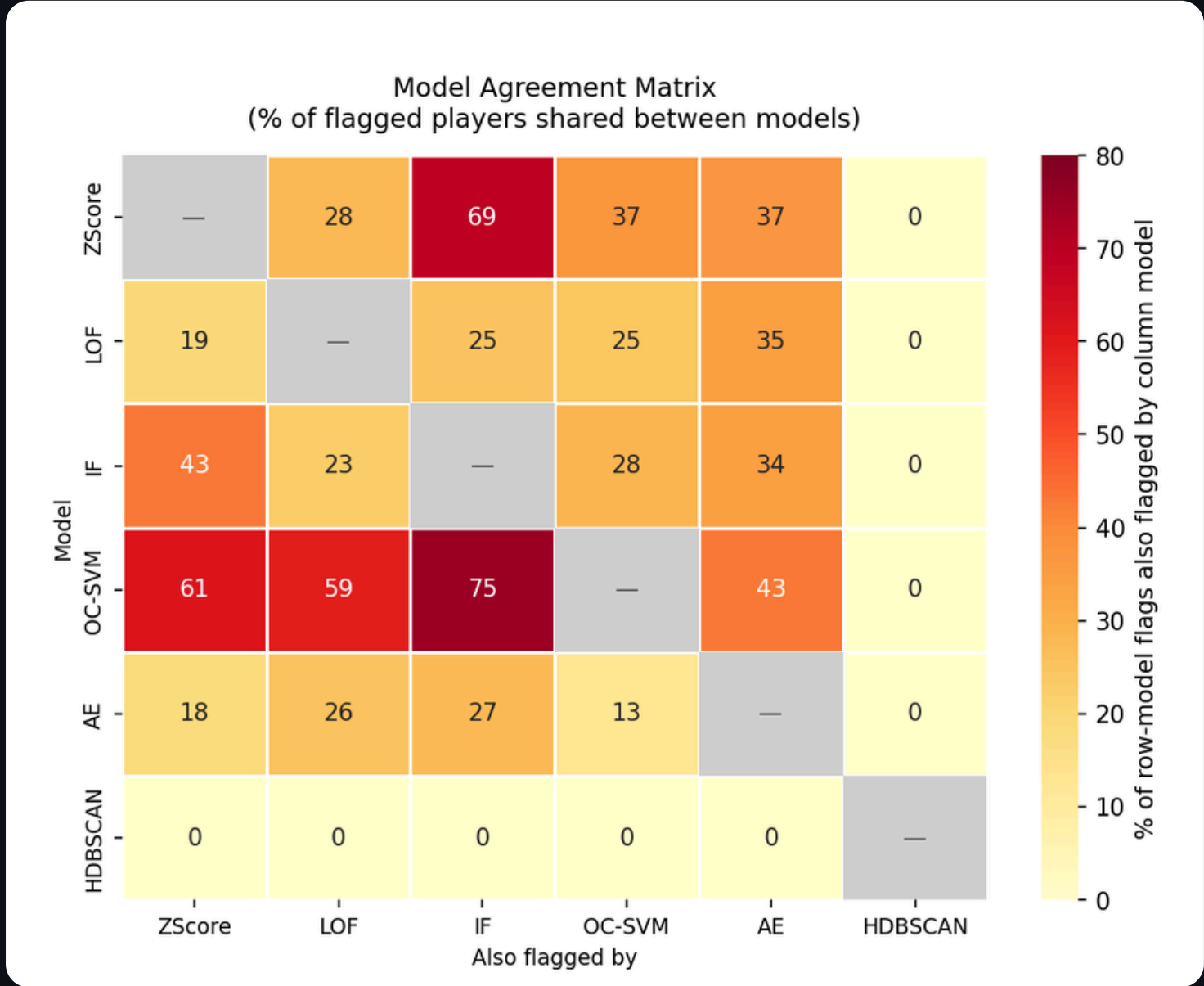
ROC curves

Diagnostics

Flagged players sit at the periphery.



UMAP: flagged players cluster at the *edge* of the manifold, not random.



Agreement: LOF + AE agree on only **26–35%** of flags.

Failure Analysis and Limitations

Realistic cheaters expose a hard ceiling.

On **realistic_cheater**, LOF Recall@k collapses to **0.01**.

These cheaters keep **behavioral features normal** so only accuracy is manipulated.

Missing signal: **move_time_cv**, engines show unnaturally low timing variance.

Requires **clock-annotated data, 2017+ Lichess solves this**.

All AUCs measure **synthetic anomalies** we designed and they are not confirmed real cheaters.

LOF · REALISTIC_CHEATER

0.01 Recall@k

CLOSES THE GAP

2016 → **2017+** Lichess data with clocks

Conclusions

A behavioral triage system that works, within known limits.

LOF: AUC 0.971 , data-efficient and peak performance achieved at just 10% of training data.

Ensemble beats any single model: complementarity proven by 26–35% flag overlap.

Behavioral > accuracy features alone:
ACPLSubAutoencoder consistently underperformed.

Hard ceiling without clock data: realistic evasion requires move-time features from 2017+ Lichess.

17,909

PLAYERS ANALYZED

21

FEATURES ENGINEERED

312

FLAGGED FOR REVIEW · 2.5%